

1 Introduction and overview

1.1 Overview

1. Linear algebra
 - Matrix equations, inverses and determinants
 - Eigenvalue problems, eigenvectors and diagonalisation
 - Vector spaces, basis vectors, inner products and norms
2. Principles of quantum mechanics:
 - States, operators, eigenvalues + measurements, Born rule
 - Hilbert space formalism
 - Time evolution, Schrödinger equation
3. Two-state and three-state systems
4. Wave mechanics: Schrödinger eqn for 1-d systems
 - Translation from general formalism
 - Potential barriers, tunneling
 - Solutions to specific equations (harmonic oscillator, hydrogen atom)
5. Harmonic oscillator
6. Bosons and fermions
7. Hydrogen atom

1.2 Introduction

There are many different ways of presenting quantum mechanics. It is quite common to begin with a historical overview of what may be called pre-quantum mechanics or quantum physics, including Planck's law of radiation, the photoelectric effect, and Bohr's model of the atom. This is done in Chapter 1 of the MP363 lecture notes by Charles Nash and will not be repeated here.

Introductory quantum mechanics is usually taught primarily through the wavefunction formalism, which you will have been introduced to last year (in EP205/MP215 or MP205). In this module, we will take the somewhat unusual approach of starting with the matrix formulation, which is the formulation used by Werner Heisenberg when he first formulated quantum mechanics in June 1925 on the island of Helgoland, where he had "escaped" to avoid serious hayfever. It is also closer to the more general, abstract Hilbert space formalism, and some of the core concepts of quantum mechanics, such as operators and eigenvalues, may be easier to grasp in this setting than in the wavefunction setting. On the other hand, wavefunctions are arguably more visualisable than vectors and matrices. Also, while wavefunctions are extremely useful in quantum mechanical systems such as atomic and solid state physics, the matrix approach is indispensable when discussing spin systems and quantum computing.

Finally, there is a practical reason for starting with the matrix approach. Whatever order we do it, we will need familiarity with some core concepts from linear algebra, and the first part of the module will be devoted to introducing or revising these concepts. It then follows naturally to continue to apply these concepts directly to quantum mechanics, and even to introduce some more advanced linear algebra ideas through quantum mechanics. That is what we will do here.

1.3 Principles of QM

We start first of all with writing down five basic principles of quantum mechanics, which may be taken as axioms or postulates that most of quantum mechanics can be derived from.

States, observables, time evolution.

2 Linear algebra

2.1 Matrices: recap

1. Matrix notation, dimensions $n \times m$ (rows $\downarrow$, columns $\rightarrow$) Row, column vectors: $1 \times m$, $n \times 1$ matrices. In physics: all elements must have the same units.
2. Multiplication rules:

$$(AB)_{ij} = \sum_k A_{ik} B_{kj} \quad (2.1)$$

This rule gives restrictions on what you can multiply, since the number of columns in A must equal the number of rows in B for (2.1) to make sense.

3. $AB \neq BA$ in general, as we can see explicitly in the following example:

$$\begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ -4 & 5 \end{pmatrix} \quad (2.2)$$

$$\begin{pmatrix} 0 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 6 & 2 \end{pmatrix} \quad (2.3)$$

The matrices used in quantum mechanics are either column vectors, row vectors, or $n \times n$ (square) matrices. An important operation is multiplying a column vector by a square matrix, which gives another column vector:

$$\begin{pmatrix} 0 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Commutator $[A, B] \equiv AB - BA$ is important in QM.

4. Transpose matrix: $(A^T)_{ij} = A_{ji}$ Important rule: $(AB)^T = B^T A^T$:

$$(B^T A^T)_{ij} = \sum_k (B^T)_{ik} (A^T)_{kj} = \sum_k B_{ki} A_{jk} = (AB)_{ji}$$

2.2 Matrices and linear equations

A set of coupled linear equations, for example 2 equations with two unknowns,

$$a_{11}x + a_{12}y = b_1 \quad (2.4)$$

$$a_{21}x + a_{22}y = b_2 \quad (2.5)$$

is equivalent to the matrix equation $AX = B$ with

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}. \quad (2.6)$$

This equation has the solution (if a solution exists) $X = A^{-1}B$, although this is often not useful in practice. The general solution in can be written in terms of the *determinant* $\det A$, and this can be generalised to $N \times N$ matrices/equations.

For the 2×2 system (2.4), (2.5) we can solve explicitly for x and y to find

$$\begin{aligned} a_{11}x + a_{12}y &= b_1 && \times a_{21}, \\ a_{21}x + a_{22}y &= b_2 && \times a_{12}, \end{aligned}$$

and subtracting these gives us

$$(a_{11}a_{22} - a_{21}a_{12})y = a_{11}b_2 - a_{21}b_1 \quad (2.7)$$

$$y = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{21}a_{12}} = \frac{a_{11}b_2 - a_{21}b_1}{\det A} \quad (2.8)$$

$$x = \frac{b_1 - a_{12}y}{a_{11}} = \frac{a_{22}b_1 - a_{12}b_2}{\det A} \quad (2.9)$$

where $\det A \equiv a_{11}a_{22} - a_{12}a_{21}$. Hence, a solution exists if $\det A \neq 0$.

We can now do the same for a 3×3 matrix (or 3 equations with 3 unknowns):

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \quad (2.10)$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \quad (2.11)$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \quad (2.12)$$

$$a_{11}(2.11) - a_{21}(2.10) : \quad (a_{11}a_{22} - a_{12}a_{21})x_2 + (a_{11}a_{23} - a_{13}a_{21})x_3 = a_{11}b_2 - a_{21}b_1 \quad (2.13)$$

$$a_{11}(2.12) - a_{31}(2.10) : \quad (a_{11}a_{32} - a_{12}a_{31})x_2 + (a_{11}a_{33} - a_{13}a_{31})x_3 = a_{11}b_3 - a_{31}b_1 \quad (2.14)$$

Denoting the terms multiplying x_2 in these two equations c_2 and c_3 respectively, we can now take $c_2(2.14) - c_3(2.13)$, and obtain

$$\begin{aligned} & [(a_{11}a_{22} - a_{12}a_{21})(a_{11}a_{33} - a_{13}a_{31}) - (a_{11}a_{32} - a_{31}a_{23})(a_{11}a_{23} - a_{21}a_{13})]x_3 \\ & = c_2(a_{11}b_3 - a_{31}b_1) - c_3(a_{11}b_2 - a_{21}b_1) \end{aligned} \quad (2.15)$$

At this point, we are only interested in whether a solution exists, which is the case if the term in square brackets on the left hand side is not zero. Expanding and simplifying it, we find

$$\begin{aligned} [] &= a_{11}^2 a_{22} a_{33} - a_{11} a_{22} a_{13} a_{31} - a_{11} a_{12} a_{21} a_{33} + a_{12} a_{21} a_{12} a_{31} \\ &\quad - a_{11}^2 a_{23} a_{32} + a_{11} a_{13} a_{32} a_{21} + a_{11} a_{12} a_{23} a_{31} - a_{12} a_{21} a_{12} a_{31} \end{aligned} \quad (2.16)$$

$$= a_{11} [a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})], \quad (2.17)$$

where we note that the last term on each line of (2.16) cancel. We identify the term in the square bracket on the last line as the determinant:

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \quad (2.18)$$

Hence, our take-home message from this is that the set of equations has a solution if $\det A \neq 0$, *unless* the right hand side is also zero. This generalises to $N \times N$ matrices/equations for any N .

If $\det A = 0$, then A is called a *singular matrix*, and its inverse A^{-1} does not exist.

Examples:

Computing the determinants of the matrices in (2.2) and (2.3), we find

$$\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = (2)(-2) - (1)(3) = -7 \quad (2.19)$$

$$\begin{vmatrix} 0 & -1 \\ 2 & 2 \end{vmatrix} = (0)(2) - (-1)(2) = 2 \quad (2.20)$$

$$\begin{vmatrix} 6 & 4 \\ -4 & -5 \end{vmatrix} = (6)(-5) - (-4)(4) = -30 + 16 = -14 \quad (2.21)$$

$$\begin{vmatrix} -1 & 2 \\ 6 & 2 \end{vmatrix} = (-1)(2) - (6)(2) = -2 + 12 = -14 \quad (2.22)$$

We can see that the last two are equal, and that they equal the product of the first two. This is an example of a general rule:

$$\det(AB) = (\det A)(\det B). \quad (2.23)$$

Example of a 3×3 determinant:

$$\begin{vmatrix} 2 & -3 & 1 \\ 1 & 3 & 2 \\ 1 & -1 & 3 \end{vmatrix} = 2(3+2) + 3(1+2) + 1(-1+3) = 10 + 9 + 2 = 21$$

2.3 Eigenvalue equations

Consider the equation $Ax = \lambda x$ where λ is a number. $X = 0$ (the null vector) is always a solution to this equation. This is called the trivial solution. Nontrivial solution is where $X \neq 0$. This is only possible for a finite set of values for λ . Corresponding X -vectors are called eigenvectors. Can rewrite $(A - \lambda I)X = 0$.

If $B = A - \lambda I$ has an inverse, then this has a solution

$$X = B^{-1} \cdot 0 = 0 \quad \Rightarrow \quad \text{only the trivial solution exists.}$$

Therefore, nontrivial solutions require

$$(A - \lambda I)^{-1} \text{ does not exist}$$

$$\Leftrightarrow \det(A - \lambda I) = 0$$

Example 1:

$$A = \begin{pmatrix} 2 & 5 \\ 1 & -2 \end{pmatrix}$$

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 5 \\ 1 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) - 5 \\ &= \lambda^2 - 4 - 5 = \lambda^2 - 9 \end{aligned}$$

$$\det(A - \lambda I) = 0 \quad \Leftrightarrow \quad \lambda^2 = 9 \quad \Rightarrow \quad \lambda = \pm 3$$

Eigenvectors:

For $\lambda = 3$:

$$\begin{pmatrix} 2 & 5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 3 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$2a + 5b = 3a \quad \Rightarrow \quad a = 5b$$

$$a - 2b = 3b \quad \Rightarrow \quad a = 5b$$

For $\lambda = -3$:

$$\begin{pmatrix} 2 & 5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -3 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$2a + 5b = -3a \quad \Rightarrow \quad a = -b$$

$$a - 2b = -3b \quad \Rightarrow \quad a = -b$$

So we have (for example):

$$\begin{pmatrix} 2 & 5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Example 2:

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = -\lambda(\lambda^2 - 1) = -\lambda(\lambda - 1)(\lambda + 1)$$

$$\Rightarrow \lambda = 0, \pm 1$$

For $\lambda = 0$:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$v_3 = 0, \quad v_2 = 0 \quad \Rightarrow \quad V = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

For $\lambda = 1$:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$v_3 = v_2, \quad v_2 = v_3 \quad \Rightarrow \quad V = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda = -1$:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = - \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$v_3 = -v_2, \quad v_2 = -v_3 \quad \Rightarrow \quad V = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

Note: $Av = \lambda v \Rightarrow A(\alpha v) = \lambda(\alpha v)$

Example 3:

$$A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\det(A - \lambda I) = (-1 - \lambda)(-1 - \lambda) - 1 = \lambda^2 + 2\lambda + 1 - 1$$

$$= \lambda^2 + 2\lambda$$

$$\lambda = \frac{2 \pm \sqrt{4 - 4(2)}}{2} = \pm i$$

Eigenvalues can be complex?!

It can be shown that the determinant of a matrix is equal to the product of its eigenvalues.

2.4 Inner product

Recap scalar product $\vec{a} \cdot \vec{b} = \sum a_i b_i = a^T b$, $|\vec{a}|^2 = \vec{a} \cdot \vec{a}$

Easily generalised to arbitrary dimensions (Lorentz relativity). With complex numbers the natural generalisation: inner product

$$\langle a|b \rangle \equiv \langle a|b \rangle = \sum a_i^* b_i, \quad a^T b \Rightarrow \langle a|a \rangle = ||a||^2 \in \mathbb{R}$$

3 Quantum mechanics: matrix formulation

3.1 Principles of quantum mechanics

In the complex matrix formulation, the principles of QM are given by

1. States $|\psi\rangle$, Normalisation $\langle\psi|\psi\rangle = 1$
2. Observables $A \rightarrow \hat{A}$ operators/matrices
3. Observed values $a_n =$ eigenvalues of $\hat{A}$
4. Expectation values $\langle A \rangle = \langle\psi|\hat{A}|\psi\rangle$
5. Time evolution $i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H|\psi\rangle$

3.2 2-state systems

Introduction to 2-state systems: spin- $\frac{1}{2}$, polarisation, isospin, qubits

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\rightarrow$ spin operators

Clearly,

$$\sigma_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \sigma_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} = - \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Let us call these $|\uparrow\rangle, |\downarrow\rangle$

If we apply a magnetic field, the energy is

$$H = -\vec{\mu} \cdot \vec{B} = -g \frac{q}{m} \vec{S} \cdot \vec{B},$$

where q is the charge of the particle. For electron, $q = -e$, and taking $\vec{B} = B\hat{z}$, we get

$$H = g \frac{e\hbar}{2m} \sigma_z$$

The two discrete spin states can be seen in the Stern-Gerlach experiment, where the spin gives rise to a deflection in the path due to a torque from magnetic field gradient. In the original experiment, silver atoms were sent through a spatially varying magnetic field, and instead of a continuous distribution, the atoms accumulated in two discrete points on the screen, corresponding to the two spin states of the atom.

Example [Exam January 2023]

In a two-state quantum mechanical system, the state $|a\rangle$ is defined as

$$|a\rangle = 2\alpha|\phi_1\rangle + i\alpha|\phi_2\rangle = \begin{pmatrix} 2\alpha \\ i\alpha \end{pmatrix},$$

where α is a real constant. The observable M is represented by the hermitean operator

$$\hat{M} = \begin{pmatrix} 1 & \sqrt{2}i \\ -\sqrt{2}i & 2 \end{pmatrix}.$$

1. Determine the value of α such that $|a\rangle$ is normalised.
2. Compute the expectation value of M in the state $|a\rangle$.
3. If M is measured, what are the possible answers that can be obtained?

Answer:

1.

$$\langle a|a\rangle = (2\alpha \quad -i\alpha) \begin{pmatrix} 2\alpha \\ i\alpha \end{pmatrix} = 4\alpha^2 + \alpha^2 = 5\alpha^2 = 1 \quad \Longleftrightarrow \quad \alpha = \frac{1}{\sqrt{5}}. \quad (3.1)$$

2.

$$\begin{aligned} \langle a|M|a\rangle &= \frac{1}{5} (2 \quad -i) \begin{pmatrix} 1 & \sqrt{2}i \\ -\sqrt{2}i & 2 \end{pmatrix} \begin{pmatrix} 2 \\ i \end{pmatrix} = \frac{1}{5} (2 \quad -i) \begin{pmatrix} 2 - \sqrt{2} \\ (-2\sqrt{2} + 2)i \end{pmatrix} \\ &= \frac{1}{5} (4 - 2\sqrt{2} - 2\sqrt{2} + 2) = \frac{2}{5} (3 - 2\sqrt{2}) \end{aligned} \quad (3.2)$$

3. The possible answers are the eigenvalues of $\hat{M}$, which are given by

$$\det(M - \lambda \mathbf{1}) = \det \begin{pmatrix} 1 - \lambda & \sqrt{2}i \\ -\sqrt{2}i & 2 - \lambda \end{pmatrix} = (1 - \lambda)(2 - \lambda) - 2 = \lambda^2 - 3\lambda = 0. \quad (3.3)$$

The possible answers are $M = 0$ and $M = 3$.

We may also find the corresponding eigenvectors.

For $\lambda = 0$:

$$v_1 + \sqrt{2}iv_2 = 0, \quad -\sqrt{2}iv_1 + 2v_2 = 0 \quad (3.4)$$

$$\Rightarrow \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2}i \\ -1 \end{pmatrix} \quad (3.5)$$

For $\lambda = 3$:

$$v_1 + \sqrt{2}iv_2 = 3v_1 \quad \Rightarrow \quad 2v_1 = \sqrt{2}iv_2 \quad \Rightarrow \quad iv_2 = \sqrt{2}v_1 \quad (3.6)$$

$$\Rightarrow \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -i\sqrt{2} \end{pmatrix}. \quad (3.7)$$

We can note that the the eigenvectors corresponding to $\lambda = 0$ and $\lambda = 3$ are orthogonal to each other.

3.3 Probabilities and the Born rule

Let us look at the expectation value of σ_z (the z -component of the spin) in the state $|a\rangle$:

$$\langle a_1 | \sigma_z | a_2 \rangle = \frac{1}{5}(2-i) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ i \end{pmatrix} = \frac{1}{5}(2-i) \begin{pmatrix} 2 \\ -i \end{pmatrix} = \frac{1}{5}(4-1) = \frac{3}{5}. \quad (3.8)$$

We can also write $|a\rangle$ as

$$|a\rangle = \frac{2}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{i}{\sqrt{5}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = c_{\uparrow} |\uparrow\rangle + c_{\downarrow} |\downarrow\rangle \quad (3.9)$$

and we see that

$$\langle a_1 | \sigma_z | a_2 \rangle = \left| \frac{\sqrt{2}}{\sqrt{5}} \right| (1) + \left| \frac{i}{\sqrt{5}} \right|^2 (-1) \quad (3.10)$$

This is an example of the *Born rule*: If

$$|\psi\rangle = \sum_n c_n |n\rangle, \hat{F}|n\rangle = f_n |n\rangle, \quad (3.11)$$

then the probability of obtaining the outcome f_n in a measurement of F is given by

$$P(f_n) = |c_n|^2. \quad (3.12)$$

We will prove this later.

3.4 Variance of an observable

The variance is defined in statistics as

$$\text{Var } F \equiv \langle (F - \langle F \rangle)^2 \rangle = \langle F^2 \rangle - \langle F \rangle^2$$

Example:

$$\hat{F} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \hat{F}^2 = \begin{pmatrix} 5 & 5 \\ 5 & 10 \end{pmatrix}$$

If $|\psi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ then $\langle \psi | \hat{F} | \psi \rangle = 2$, $\langle \psi | \hat{F}^2 | \psi \rangle = 5$.

$$\text{Var } F = \langle F^2 \rangle - \langle F \rangle^2 = 5 - 2^2 = 1$$

Uncertainty/spread:

$$\Delta F \equiv \sqrt{\text{Var } F} = \sqrt{1} = 1$$

3.5 Properties of Hermitian matrices and inner products

(a) $\langle U | V \rangle = \sum u_i^* v_i = (\sum u_i v_i^*)^* = \langle V | U \rangle^*$

(b)

$$\begin{aligned} \langle U | M | V \rangle &\equiv \langle U | M V \rangle = \sum_{i,j} u_i^* M_{ij} v_j = \sum_{i,j} (M_{ij} u_i^*) v_j \\ &= \sum_{i,j} [(M^T)_{ji} u_i^*] v_j = \sum_{i,j} [(M^T)_{ji}^* u_i]^* v_j = \sum_j (M^\dagger u)_j^* v_j \\ &= \langle M^\dagger U | V \rangle = \langle U | M^\dagger V \rangle \end{aligned} \quad (3.13)$$

(c) If $H = H^\dagger$ then

$$\langle \psi | H | \psi \rangle = \langle \psi | H^\dagger | \psi \rangle = \langle \psi | H | \psi \rangle^* \quad (3.14)$$

$$\Rightarrow \langle \psi | H | \psi \rangle \in \mathbb{R} \quad \text{for any state } |\psi\rangle \quad (3.15)$$

(d) If $\hat{F}|m\rangle = f_m|m\rangle$ and $\hat{F}|n\rangle = f_n|n\rangle$ then

$$\langle m | \hat{F} | n \rangle = f_n \langle m | n \rangle \quad (3.16)$$

$$= \langle n | \hat{F}^\dagger | m \rangle^* = \langle n | \hat{F} | m \rangle^* = f_m \langle n | m \rangle^* = f_m \langle m | n \rangle \quad (3.17)$$

$$\Rightarrow \langle m | n \rangle = 0 \quad \text{if } f_m \neq f_n. \quad (3.18)$$

This means that the eigenvectors of $\hat{F}$ form an *orthogonal set*.

3.6 Hermitean operators, basis vectors, and completeness

We found that eigenvectors $|n\rangle$ of a Hermitian operator $\hat{F}$ are mutually orthogonal, $\langle m|n\rangle = 0$ if $f_m \neq f_n$.

If $|m\rangle$ are all normalised they form an *orthonormal* set.

If $f_m = f_n$ but $|n\rangle \neq |m\rangle$ we can *make* them orthogonal (we will not prove this here).

In an N -dimensional space, $\hat{F}$ will have N eigenvalues and corresponding eigenvectors.

We can *choose* these to be our basis vectors and express all others in terms of those. Compare 2- or 3-dimensional space: *any* set of mutually orthogonal unit vectors can be chosen to define our x, y, z axes.

$$\begin{aligned} |\psi\rangle &= \sum_n c_n |n\rangle, \quad \langle m|\psi\rangle = \sum_n c_n \langle m|n\rangle = \sum_n c_n \delta_{mn} = c_m \\ \Rightarrow |\psi\rangle &= \sum_n |n\rangle \langle n|\psi\rangle \quad (\text{completeness relation}) \quad \sum_n |n\rangle \langle n| = 1 \end{aligned}$$

Example (from L7):

$$\begin{aligned} |\varphi_1\rangle &= \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix}, \quad |\varphi_2\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} \\ |\psi\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ \sqrt{2} \end{pmatrix} \\ \langle \varphi_1|\psi\rangle + \langle \varphi_2|\psi\rangle &= \frac{1}{3} \begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & -1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & \sqrt{2} \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ |\psi\rangle &= \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix} \\ \langle \varphi_1|\psi\rangle &= \frac{1}{3} (\sqrt{2} \cdot \sqrt{2} + (-1) \cdot 1) = \frac{1}{3} (2 - 1) = \frac{1}{3} \\ \langle \varphi_2|\psi\rangle &= \frac{1}{3} (1 \cdot \sqrt{2} + \sqrt{2} \cdot 1) = \frac{2\sqrt{2}}{3} \\ |c_1|^2 + |c_2|^2 &= \frac{1}{9} + \frac{8}{9} = 1 \end{aligned}$$

This is again an example of Born rule probabilities for different outcomes.

Born rule proof

We can now prove the Born rule in the general case:

$$\begin{aligned} \langle F \rangle &= \langle \psi | \hat{F} | \psi \rangle = \left(\sum_m c_m^* \langle m | \right) \hat{F} \left(\sum_n c_n | n \rangle \right) = \sum_{mn} c_m^* c_n \langle m | \hat{F} | n \rangle \\ &= \sum_{mn} c_m^* c_n F_{mn} \langle m | n \rangle = \sum_n c_n^* c_n F_{nn} = \sum_n |c_n|^2 F_{nn} \end{aligned}$$

3.7 Time evolution, Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

is in general a solvable matrix differential equation $\Rightarrow N$ coupled differential equations.

Assume $\hat{H} |\psi(0)\rangle = E |\psi(0)\rangle$, $|\psi(0)\rangle$ is an energy eigenstate.

Then:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = E |\psi(t)\rangle$$

So maybe:

$$|\psi(t)\rangle = e^{-iEt/\hbar} |\psi(0)\rangle$$

is a solution? Check: Yes.

Now take:

$$\hat{H} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad |\psi_0\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Write $|\psi(t)\rangle = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}$.

$$\hat{H}|\psi\rangle = i\hbar \frac{\partial}{\partial t}|\psi\rangle \Rightarrow \begin{cases} au(t) = i\hbar \dot{u}(t) \\ bv(t) = i\hbar \dot{v}(t) \end{cases}$$

$$u(t) = e^{-iat/\hbar}u_0, \quad v(t) = e^{-ibt/\hbar}v_0$$

This holds for any $\hat{H}$ that is diagonal — but $\hat{H}$ is always diagonal if we choose the energy eigenstates as the basis.

Equivalently:

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle \quad \Rightarrow \quad |\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle$$

3.8 Commutation relations

Two observables F and G can be measured simultaneously iff all eigenstates of F are also eigenstates of G , i.e.,

$$\hat{F}|f_n\rangle = f_n|f_n\rangle, \quad \hat{G}|g_n\rangle = g_n|g_n\rangle$$

But then:

$$\begin{aligned} \hat{F}\hat{G}|\psi\rangle &= \sum_n \hat{F}\hat{G}c_n|n\rangle = \sum_n c_n \hat{F}g_n|n\rangle = \sum_n c_n g_n f_n|n\rangle \\ &= \sum_n g_n \hat{F}|n\rangle = \hat{G}\hat{F}|\psi\rangle \end{aligned}$$

Example:

$$\hat{F} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \hat{G} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{F}\hat{G} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad \hat{G}\hat{F} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \quad \Rightarrow \quad [\hat{F}, \hat{G}] = 2 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

3.9 Collapse of wavefunction and measurement problem

In standard textbook mechanics, the Schrödinger equation only governs the evolution of the quantum state *between* measurements, or between the preparation of an initial state and the measurement of the properties of the system. As we know, any measurement will always give a definite outcome, and at this point we know for definite that the system has the corresponding property. For example, a spin- $\frac{1}{2}$ particle can be prepared in a superposition of its two spin states, but if it is then sent through a Stern–Gerlach experiment it will deflect one way or the other and not both at the same time. If this deflection corresponds to the spin-up state, we can conclude after the measurement that the particle is in the spin-up state. This means that

At the point of measurement, the state goes from being a superposition of eigenstates of the measured quantity to being in the eigenstate corresponding to the measured value.

This is called the *projection postulate* (after John von Neumann, who first formulated it in 1932), or *reduction of the state vector*, or, more popularly, *collapse of the wave function*. This is one of the most contentious issues in the formulation and interpretation of quantum mechanics.